

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Digital Signal Analysis and Processing

Semester: Fall

Year : 2021

Full Marks: 100

Pass Marks: 45

Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

- a) Define Signal Analysis and Signal Processing. What are the advantages of Digital signal processing over Analog signal processing? 8 1
- b) The output of an LTI system is $y(n) = \{2, 5, 1, \underline{-10}, -10, -3, 6, 9\}$ to an impulse response $h(n) = \{1, \underline{2}, 0, -3\}$. Find the input $x(n)$ using convolution sum. The underline represents the element at $n = 0$. 7 2.5
- a) Find the convolution between two signals
 $x[n] = u[n]$ and $h[n] = a^n u[n]$, $0 < a < 1$ 7 2
- b) Determine Z-transform along with ROC of
$$x(n) = \left[\left(\frac{1}{2}\right)^n + \left(\frac{3}{4}\right)^n \right] u(n-5)$$
 8 3
3. a) Find the 8-point DITFFT of $x(n) = \sin \frac{3\pi n}{4}$, for $0 \leq n \leq 7$ 8 4
- b) What is Zero Padding? Find the circular Convolution of $x_1[n] = \{1, 2\}$ & $x_2[n] = \{3, 2, 1\}$ 7 4
4. a) Obtain the parallel form realization of following IIR filter. 7 5
- $$H(z) = \frac{3(2z^2 + 5z + 4)}{(2z + 1)(z + 2)}$$
- b) Compute Lattice coefficients and draw lattice structure for given IIR system. Also check stability of the system. 8 5
- $$H(z) = 1 / (1 - 0.525z^{-1} + 0.6125z^{-2} + 0.3z^{-3})$$
5. a) Design a lowpass filter which will have -3 dB cut off at 30π rad/sec and an attenuation of 50 dB at 45π rad/sec. The filter is required to have a linear phase and the system uses a sampling rate of 100 samples/sec. 8 6.4
- b) Design a linear FIR filter using Kaiser window to meet the following specifications: 7 6.3

$$0.99 \leq |H(e^{j\omega})| \leq 1.01, \text{ for } 0 \leq |\omega| \leq 0.19\pi$$

$$\leq |H(e^{j\omega})| \leq 0.01, \text{ for } 0.21\pi \leq |\omega| \leq \pi$$

6. a) Determine $H(z)$ using impulse invariance method for the following analog filter's transfer function :

$$H_a(s) = \frac{1}{(s + 0.5) + (s^2 + 0.5s + 2)}$$

- b) Design a digital lowpass Butterworth filter using Bilinear Transformation method to meet the following specifications:

Passband attenuation = 1 dB

Stopband attenuation = 16 dB

Passband edge = 120 Hz

Stopband edge = 170 Hz

Sampling frequency = 256 Hz

7. Write short notes on: (Any two)

- Region of Convergence
- Symmetric and Anti-symmetric filter
- Kaiser window